By Theorem 1, $\det \begin{bmatrix} 3 & -7 & 8 & -6 \\ 0 & 2 & -5 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{bmatrix} = a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} + a_{41}C_{41}$

We know, $a_{21} = a_{31} = a_{41} = 0$.

$\therefore a_{11}C_{11} = 3 \cdot (-1)^{1+1} \det \begin{bmatrix} 2 & -5 & 3 \\ 0 & 1 & 0 \\ 0 & 2 & -1 \end{bmatrix} = 3 \cdot \det \begin{bmatrix} 2 & -5 & 3 \\ 0 & 1 & 0 \\ 0 & 2 & -1 \end{bmatrix}$

By Theorem 1, $\det \begin{bmatrix} 2 & -5 & 3 \\ 0 & 1 & 0 \\ 0 & 2 & -1 \end{bmatrix} = a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31}$

$a_{21} = a_{31} = 0$

$C_{11} = (-1)^{1+1} \begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} = 1(-1) - 0(2) = -1$

$\therefore a_{11}C_{11} = 2 \cdot (-1) = -2.$

$\det A = 2 \cdot 3 \cdot (-2) = -12$ ∎

The method in Example 3 can be easily adapted to prove the following theorem:

Theorem 2: If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A.

PRACTICE PROBLEM

Compute $\begin{vmatrix} 5 & -7 & 2 & 2 \\ 0 & 3 & 0 & -4 \\ -5 & -8 & 0 & 3 \\ 0 & 5 & 0 & -6 \end{vmatrix}$

Soln: By Theorem 1, $a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} + a_{43}C_{43}$ = given determinant.

$a_{23} = a_{33} = a_{43} = 0.$

$a_{13}C_{13} = 2 \cdot (-1)^{1+3} \begin{vmatrix} 0 & 3 & -4 \\ -5 & -8 & 3 \\ 0 & 5 & -6 \end{vmatrix}$

By Theorem 1, $a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31}$ = given determinant. $a_{11} = a_{31} = 0.$

$a_{21}C_{21} = -5 \cdot (-1) \begin{vmatrix} 3 & -4 \\ 5 & -6 \end{vmatrix} = 5(3 \cdot (-6) - (-4) \cdot 5)$

$= -10.$

$2 \cdot (-10) = -20$ ∎